

Exact Moments and Cumulants of Twenty Frozen First-Passage Times

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Abstract

The C88 frozen sixteen-label model supplies exact first-passage distributions for twenty subgroup targets. We compute their raw, falling-factorial, central, and cumulant moments through order six. Every entry is a rational number. A second representation reconstructs the laws from the complete C88 hit bitsets, while survival-tail summation independently recovers both ordinary and factorial moments. Probability-generating functions, clean replay, and thirteen hostile mutations complete the finite audit. The scope is `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Frozen random variables

Let L be the sixteen named labels and let A_k be the first k entries of a uniformly random permutation. For each actual frozen subgroup H define $T_H = \min\{k : H \leq \Phi(A_k)\}$. C88 records the exact count $N_H(k)$ of permutations with $T_H = k$.

2 Moment theorem

Theorem 1. *For every one of the twenty targets and $0 \leq r \leq 6$, the certificate contains the exact values*

$$m_r = \mathbb{E}[T_H^r], \quad f_r = \mathbb{E}[(T_H)_r], \quad \mu_r = \mathbb{E}[(T_H - \mathbb{E}T_H)^r],$$

and cumulants $\kappa_1, \dots, \kappa_6$. If $S_H(k) = \Pr(T_H > k)$, then

$$m_r = \sum_{k=0}^{15} ((k+1)^r - k^r) S_H(k), \quad f_r = r! \sum_{k=0}^{15} \binom{k}{r-1} S_H(k) \quad (r \geq 1).$$

Proof. The first two identities are the finite tail-sum formulas for a nonnegative integer-valued variable. Raw moments determine central moments and cumulants through the standard binomial and moment-cumulant recursions. The producer evaluates the left sides from C88 permutation counts; the independent checker evaluates the same quantities from C88's support bitsets. $\square$

3 Inventory and audit

The canonical receipt has twenty rows and 17 distribution cells per row, then 7 raw, factorial, and central cells and 6 cumulants per target. The mean in every row equals its C88 expectation; the top target is 36499/3960. SymPy verifies normalized generating functions, derivatives at one, exact central moments, and both tail identities. A clean subprocess replay preserves evidence digest

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No arithmetic/local-data, Euler-factor, root-number, automorphy, full Burnside/table-of-marks, or Hilbert–Polya operator claim is made.